Clustering agent representation for explaining reinforcement learning policies

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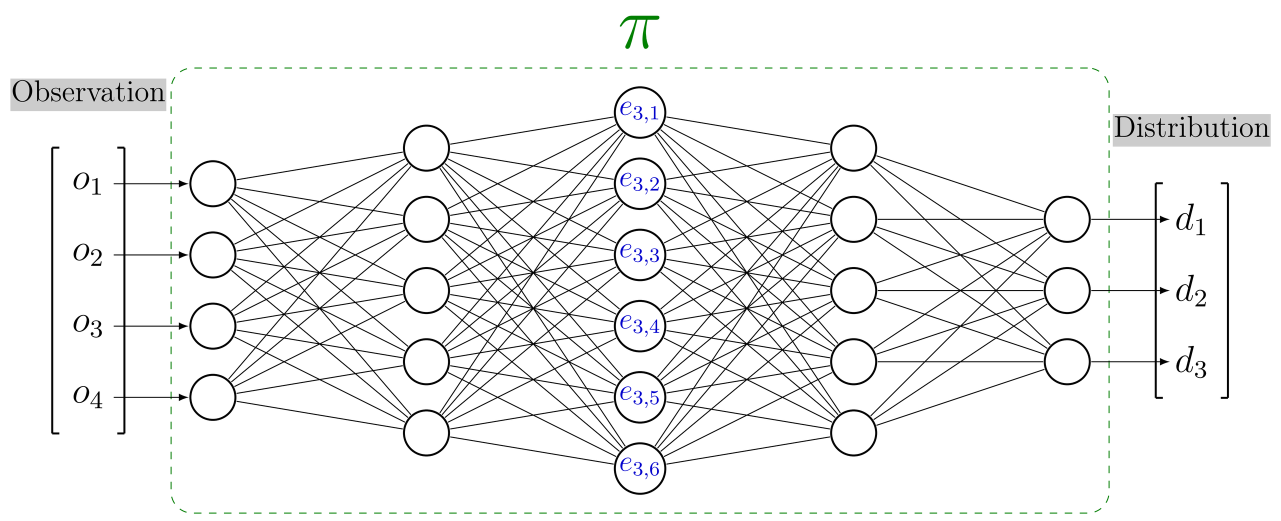
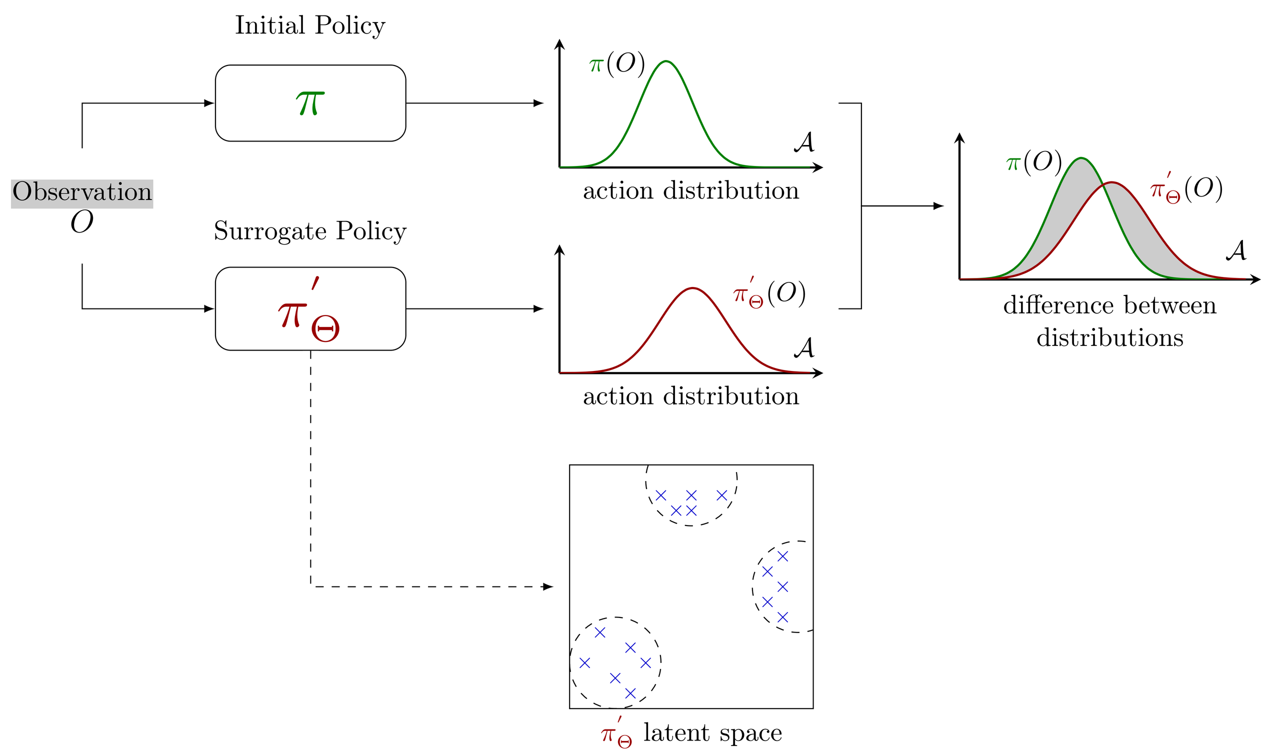
(1) Latent Space

Deep learning policies are represented as neural networks. Our goal is to study the neural network's latent space. This allows us to extract the semantics used for explainability.



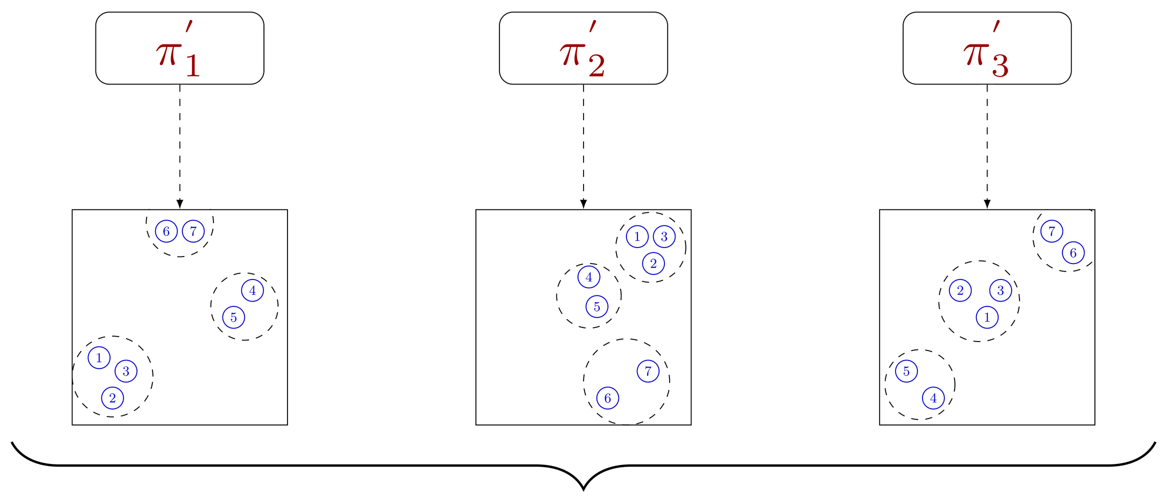
(3) Clustering Agent Representation

To ensure the representation is relevant for analysis, we repeat the latent space extraction process multiple times. This will allow us to confirm its stability before conducting comparisons between them.



(4) Clustering Universality

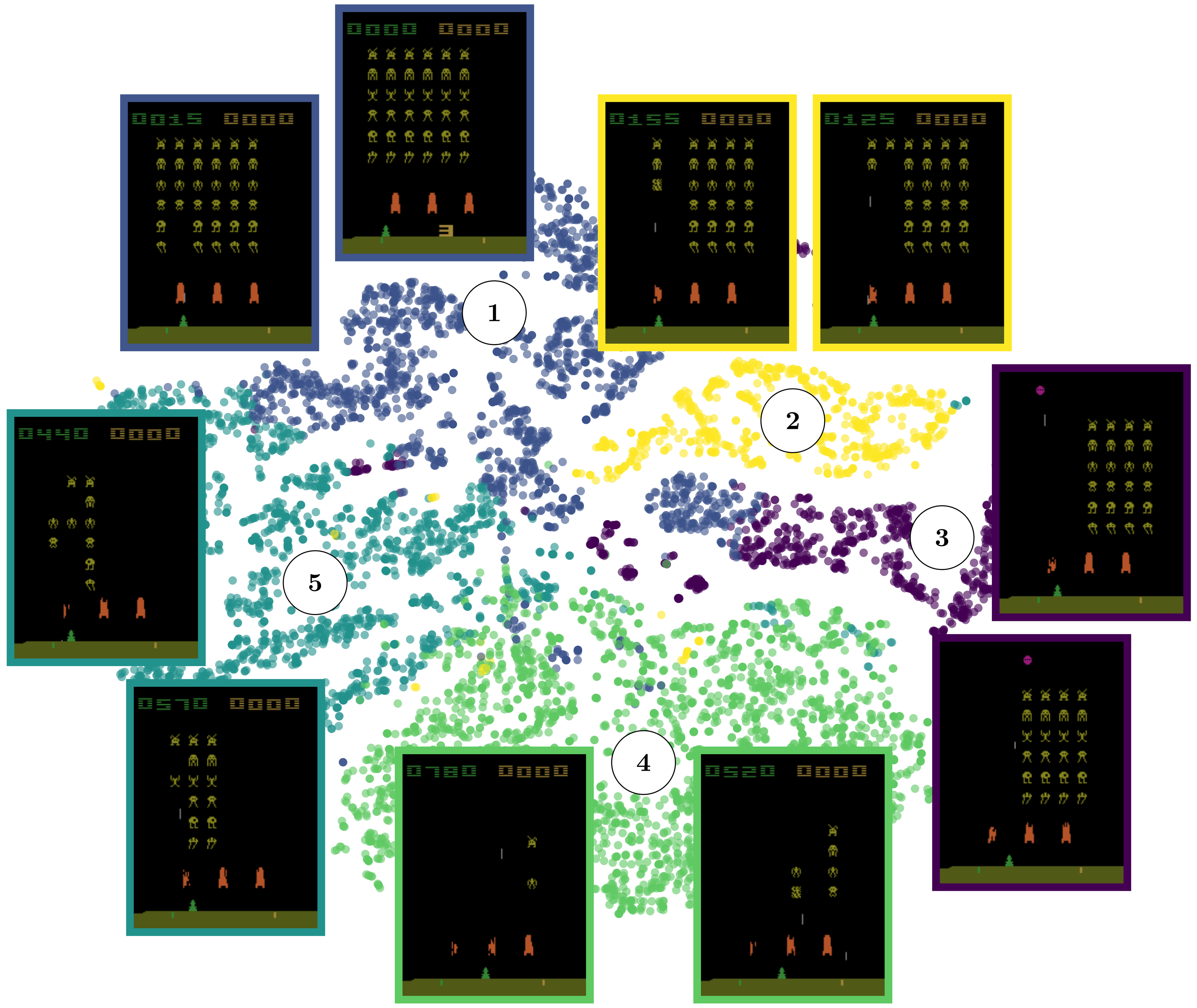
To compare whether latent spaces are similar, we use clustering analysis. Each latent space is processed with a clustering algorithm using N clusters. The Adjusted Rand Index is then used to measure the agreement between the clusterings, providing us with a stability metric.



$$\textit{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^{n} \sum_{j=0}^{i-1} \mathrm{ARI}(\mathcal{C}_i, \mathcal{C}_j)$$

(2) Semantic Latent Space

We observe that the distances between embeddings in the latent space are meaningful. The closer the embeddings are in the latent space, the closer they are semantically.



(5) Evolution Clustering Universality

We observe that regardless of the number of clusters chosen, the clustering remains universally good, with an Adjusted Rand Index of approximately 0.5. This indicates that the representations are stable across different learning processes.